

ANDREWS AFB, MD, USA

WMO: 745940

Lat: 38.817N

Lon: 76.867W

Elev: 282

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	18.1	-2.5	4.9	17.4	2.6	6.4	22.2	28.3	31.6	26.4	32.2	10.3	320	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	93.7	74.8	91.0	74.1	88.5	73.0	78.0	88.5	76.7	86.4	75.4	84.1	9.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	133.4	81.7	73.8	127.5	80.4	72.9	123.4	79.6	41.5	88.3	40.2	86.3	39.0	84.2	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	20.8	18.5	7.2	97.0	5.3	2.9	3.3	99.1	0.2	100.8	-2.8	102.4	-6.7	104.5		
			6.1	80.8	4.9	2.5	2.6	82.6	-0.3	84.1	-3.0	85.5	-6.6	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	34.4	37.6	45.2	56.0	64.7	73.3	77.6	75.8	69.3	58.0	47.3	39.4
	DBStd	16.99	10.63	9.77	9.75	8.79	7.79	6.03	5.05	4.92	6.47	8.07	8.37	9.06
	HDD50	1627	498	364	213	35	1	0	0	0	0	18	148	350
	HDD65	4348	948	767	617	298	105	10	1	0	30	246	532	794
	CDD50	4057	15	17	65	214	458	698	856	799	579	267	68	21
	CDD65	1300	0	0	3	26	97	258	391	334	159	30	2	0
	CDH74	11640	0	2	36	293	883	2288	3836	2900	1165	224	11	2
	CDH80	4371	0	0	5	81	272	834	1667	1101	363	48	0	0
Wind	WSAvg	8.4	9.8	9.8	10.4	9.6	8.0	7.5	7.0	6.6	7.0	7.6	8.5	8.7
Precipitation	PrecAvg	44.6	3.1	2.6	3.9	3.6	4.2	4.5	4.3	4.0	3.9	3.6	2.8	3.6
	PrecMax	64.8	6.1	7.1	8.9	5.9	10.0	10.7	7.1	10.1	10.9	8.5	6.2	6.6
	PrecMin	35.0	1.5	0.5	0.0	1.9	1.3	1.3	1.2	1.5	0.1	0.3	0.4	1.5
	PrecStd	6.4	1.1	1.4	1.7	1.2	2.1	2.4	1.6	1.9	2.6	2.1	1.6	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	70.9	78.8	87.8	90.5	94.6	98.0	96.9	92.4	85.1	75.2	69.5
		MCWB	60.5	58.6	60.7	67.6	72.9	75.4	76.4	75.0	72.4	69.4	62.9	61.9
	2%	DB	62.6	63.8	72.3	81.8	86.5	90.9	94.7	92.3	88.3	80.1	69.7	63.7
		MCWB	57.4	54.6	58.1	63.7	71.1	74.3	75.8	74.9	71.8	67.8	60.3	57.9
	5%	DB	57.0	58.1	66.9	76.8	82.8	88.1	91.4	89.1	84.0	75.3	65.5	58.8
		MCWB	51.2	49.1	55.6	62.2	69.1	72.9	74.6	73.6	70.2	65.3	57.8	53.3
	10%	DB	50.3	52.8	61.8	72.0	79.1	85.2	88.8	86.1	80.6	71.8	61.3	53.7
		MCWB	44.8	45.7	52.5	59.5	67.0	71.7	73.8	72.6	69.2	63.8	54.1	48.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.9	61.8	64.1	69.7	75.5	78.3	80.6	79.4	76.8	73.4	67.1	64.3
		MCDB	65.2	69.1	73.3	79.5	86.1	89.0	91.9	89.7	84.7	79.4	70.7	66.9
	2%	WB	58.0	56.3	60.5	66.9	73.2	76.5	78.4	77.5	75.0	70.5	63.3	59.4
		MCDB	62.3	62.3	69.0	77.2	83.0	86.7	89.5	87.3	82.5	76.1	66.7	63.1
	5%	WB	52.1	50.9	57.4	64.2	71.2	75.1	77.1	76.0	73.5	68.1	59.5	54.5
		MCDB	56.1	55.7	64.9	73.8	80.4	84.5	87.2	84.6	79.9	73.8	63.4	57.9
	10%	WB	45.7	46.2	53.4	61.4	69.2	73.7	75.8	74.7	72.0	64.8	55.9	49.0
		MCDB	48.9	52.0	61.0	69.7	76.2	82.1	84.9	82.2	77.8	70.2	60.1	52.6
Mean Daily Temperature Range	5% DB	MDBR	15.6	17.3	18.9	20.8	19.6	18.9	18.6	18.4	18.2	19.1	18.2	15.8
		MCDBR	22.5	24.7	26.4	27.3	24.1	22.1	21.8	21.5	22.3	23.0	22.5	21.3
		MCWBR	17.7	17.1	16.1	13.5	11.5	9.2	7.7	7.8	9.2	12.1	14.9	17.0
	5% WB	MCDBR	21.1	21.1	23.6	23.5	21.4	19.7	19.0	18.0	17.4	18.6	18.1	18.9
		MCWBR	18.5	17.4	17.2	13.7	11.0	9.3	7.9	7.8	9.1	12.0	14.8	17.2
Clear-Sky Solar Irradiance	taub	0.305	0.318	0.352	0.394	0.434	0.488	0.511	0.504	0.418	0.357	0.321	0.307	
	taud	2.501	2.450	2.415	2.314	2.231	2.122	2.076	2.079	2.294	2.470	2.530	2.534	
	Ebn at Noon	278	288	288	280	270	254	247	245	262	270	269	269	
	Edh at Noon	25	30	34	40	44	49	51	50	38	29	24	23	
All-Sky Solar Radiation	RadAvg	667	934	1242	1572	1775	1961	1921	1713	1394	1046	759	586	
	RadStd	54	103	94	118	201	149	96	92	158	115	64	55	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (78 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+137	

Nomenclature: See separate page